



CHROMATOGRAPHY

Student Learning Outcomes (SLOs)

After completing this lesson, the student will be able to:

- Describe how paper chromatography is used to separate mixture of soluble substances, using a suitable solvent.
- Describe the use of locating agents when separating mixtures containing colourless substances. (For context, knowledge of specific locating agents is not required)
- Interpret simple chromatograms (for context, students should identify:
 - a) unknown substance with known substances
 - b) pure and impure substances.
- State and use the equation for R_f .

INTRODUCTION

The word chromatography has been originated from two words one is Chroma means colour the other graphein means writing.

Chromatography has been originated as a very strong analytical tool in the analytical chemistry. This is not only used to analyse a compounds, it is used for separation of the compounds as well. Nowadays this technique can be coupled with other techniques like mass spectrometry.

19.1 PRINCIPLE OF CHROMATOGRAPHY

Chromatography works on the principle of partitioning where the components of a mixture distribute themselves between two phases one is called stationary phase and the other one is said to be mobile phase.

Stationary phase:

Stationary phase in chromatography is a solid phase or a liquid phase coated on the surface of a solid phase. Like in paper chromatography water molecules are adsorbed on paper material cellulose.

Mobile phase:

It is a liquid or gas that moves through or over the stationary phase.

As the mixture is carried by the mobile phase through the stationary phase, each component interacts differently with the two phases, leading to different migration rates. This results in the separation of the components based on their affinity for the stationary phase relative to the mobile phase.

19.2 Paper chromatography

In paper chromatography the stationery phase is the water absorbed on filter paper. The mobile phase can be a solvent or a mixture of solvents. A mixture of inks or some metal ions can be separated with the help of paper chromatography.

Procedure

1. Take 1 dm³ measuring cylinder fitted with the rubber bung carrying a glass hook at the lower end
2. Pour 25 cm³ of water and ethanol solvent mixture (6:4) in it and close it. Allow it to stand for 15 minutes.
3. Take a strip of paper (what Mann filter paper number one) 2.5 cm wide and lengthier than the length of the cylinder.
4. Mark a line 5 centimetre from one end its mid point with pencil.
5. Prepare a mixture of various inks like blue, black, green, red etc by taking two drops of each in a clean China dish. Mix them thoroughly.
6. Apply a very tiny drop of this mixture at the midpoint by means of capillary tube.
7. Dry the strip in the air for about 15 minutes and suspend it by the glass hook with the impregnated end dipping into the solvent to a depth of 5-6 mm.

8. When the solvent front has risen for about an hour or the solvent front is 2-3cm below the upper end of the filter paper.
9. Remove the paper strip and dry it in air. It is seen that every ink gives coloured bands at different regular intervals.
10. Measure the distance of these coloured bands from the baseline and the distance travelled by the solvent front from the baseline.
11. Calculate the R_f values for various inks by the formula given. In this way we can separate a mixture of various inks by chromatography.

$$R_f = \frac{\text{Distance traveled by solute}}{\text{Distance traveled by solvent}}$$

R_f factor (retardation factor) is a measure of affinity of a component with mobile phase or stationary phase. If R_f value is greater it means that the component is more affine towards the mobile phase than to stationary phase.

Locating agent

A locating agent is a substance or a solution which is used to locate the components which have been separated by chromatography but are not visible due to being colourless. For example, a mixture of amino acids can be separated by paper chromatography or by thin layer chromatography (TLC) but they are not visible, to locate them ninhydrin is sprayed on the chromatogram. Ninhydrin reacts with amino acids and develops into a colourful chromatogram.

Reading a chromatogram

Measure the distance each compound spot travelled from the starting point (origin) to its current position. This is known as the R_f (Retention Factor) value and is calculated as the distance travelled by the compound divided by the total distance travelled by the solvent. Compare the R_f values of the separated compounds to known standards data to identify the components within your chromatogram.

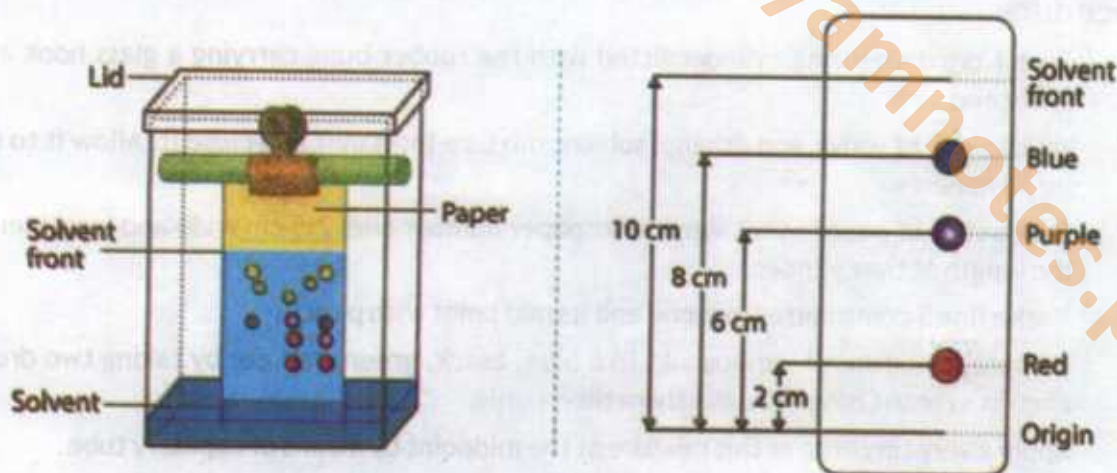


Figure 19.1: Paper chromatography

For example, in the above given chromatogram solvent front travelled 10cm and the blue coloured component travelled 8 cm, so the R_f is:

$$R_f (\text{blue}) = 8/10 = 0.8$$

$$R_f (\text{purple}) = 6/10 = 0.6$$

$$R_f (\text{red}) = 2/10 = 0.2$$

Blue component has the greatest affinity with the mobile phase and the red has the greatest affinity with stationary phase.

Identification of an unknown substance

A paper chromatogram is used to identify an unknown substance by comparing it with known substances. Two substances are likely to be same, if their R_f value is same. Therefore, R_f value of an unknown substance is determined. This R_f value is then cross-matched with already known R_f values to identify the potential substance.

A pure substance produces only one spot on the paper chromatogram. An impure substance can produce more than one spots depending on the compounds present in it.

KEY POINTS

- Chromatography works on the principle of partitioning where the components of a mixture distribute themselves between two phases one is called stationary phase and the other one is said to be mobile phase.
- In paper chromatography the stationary phase is the water absorb on filter paper
- A locating agent is a substance or a solution which is used to locate the components which have been separated by chromatography but are not visible due to being colourless
- R_f (Retention Factor) value is calculated as the distance travelled by the compound divided by the total distance travelled by the solvent

REVIEW QUESTIONS

1. Give short answer.
 - (i) Define chromatography
 - (ii) Define locating agent
 - (iii) What is R_f value?
 - (iv) Define paper chromatography
 - (v) What do you mean by paper chromatogram?
2. Explain in detail paper chromatography.
3. How can you identify an unknown substance by chromatography?
4. Differentiate between stationary and mobile phase.